

Jurisdictional Reward Funds in global equilibrium: the case of fossil fuels

July 2, 2026

<i>good with global externalities</i>	<i>substitute</i>	<i>global funds for supply reduction</i>	<i>global funds for demand reduction</i>	<i>global funds for substitute expansion</i>
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The effects of **oil** supply reductions per tCO₂

- avoided climate damages = $(1 - \text{leakage})SCC = 0.78 \times \$200 = \$156$
- terms-of-trade effects:
\$127 utilitarian welfare loss (assuming log utility)
- geopolitical effects: increased revenues for Russia $\implies$ increased spending on aggression
\$60 physical destruction (given \$25 terms-of-trade gains for Russia (Beaufils et al. 2025))
\$39 loss of life
\$21 casualty losses
- $\implies$ net welfare effect of supply reduction = $\$156 - \$127 - \$60 - \$39 - \$21 = -\91
- $\implies$ net welfare effect of demand reduction = $\$44 + \$127 + \$60 + \$39 + \$21 = \403

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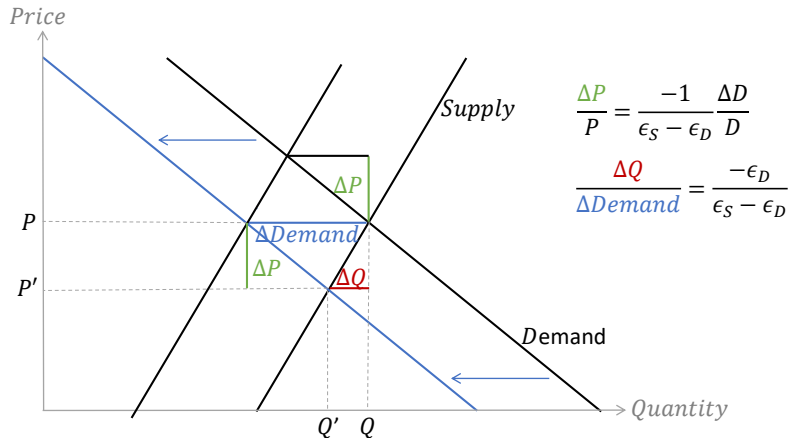
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The model

- $I =$ set of countries of the world, all price takers
- country i chooses:
 - x_i : coal extraction
 - y_i : energy use
 - z_i : renewable energy production
- global market for coal, price p
- $x_i - y_i + z_i =$ country i 's net export of coal
- $U_i(x_i, y_i, z_i) = \underbrace{B_i(y_i)}_{\substack{B'_i > 0 \\ B''_i < 0}} - \underbrace{G_i(z_i)}_{\substack{G'_i > 0 \\ G''_i > 0}} - \underbrace{C_i(x_i)}_{\substack{C'_i > 0 \\ C''_i > 0}} + p(x_i - y_i + z_i) + f_i(x_i, y_i, z_i)$

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The global institution's problem

- Global institution announces reward payment schemes:
 $f_{ix}(x_i), f_{iy}(y_i), f_{iz}(z_i) \geq 0$
- Countries choose their (x_i, y_i, z_i)

Definition

A **market equilibrium** under a given set of reward payment schemes $(f_{ix}, f_{iy}, f_{iz})_{i \in I}$ is a combination of an allocation $(x_i, y_i, z_i)_{i \in I}$ and a price p such that:

1) market clearing: $\sum_{i \in I} x_i - y_i + z_i = 0$

2) individual rationality:

$$x_i = \operatorname{argmax}_x px - C_i(x) + f_{ix}(x)$$

$$y_i = \operatorname{argmax}_y -py + B_i(y) + f_{iy}(y)$$

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Viewing the global institution as choosing the allocation and the price

market equilibrium
(surjective map)

$$(f_{xi}, f_{yi}, f_{zi})_{i \in I} \longrightarrow (p, (x_i, y_i, z_i)_{i \in I})$$

minimal required
transfers

$$(F_{ix}, F_{iy}, F_{iz})$$

$$F_{ix} = \sup_x px - C_i(x) - (px_i - C_i(x_i))$$

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- Objective: $\sum_{i \in I} B_i(y_i) - C_i(x_i) - G_i(z_i) - \eta(\sum_{j \in I} x_j)$
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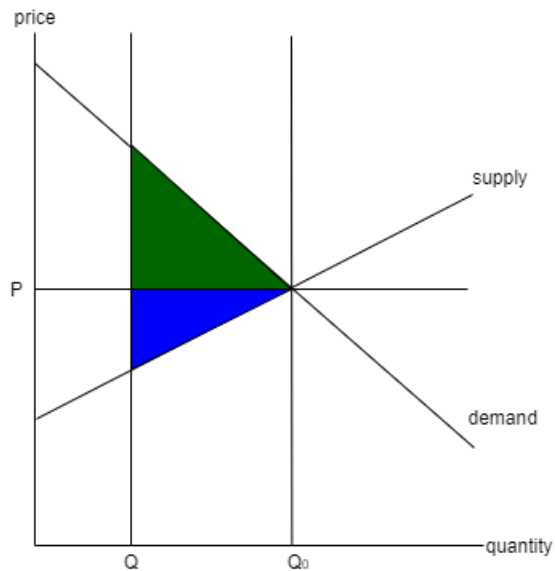
The Price Preservation Lemma

Lemma

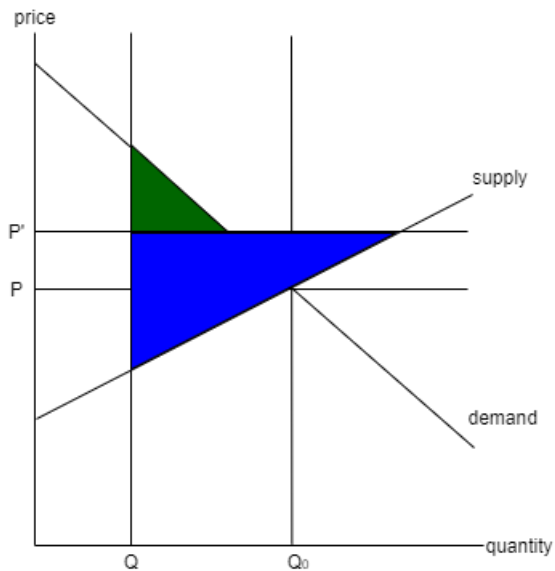
At the optimal mechanism:

The world market price p for coal ends up being the same as in the absence of any mechanism.

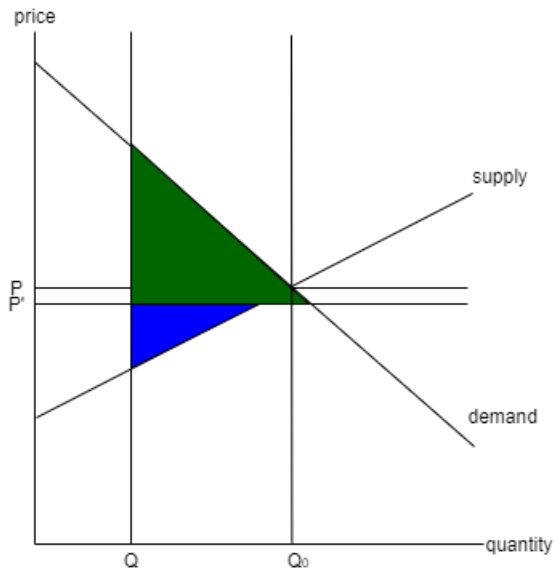
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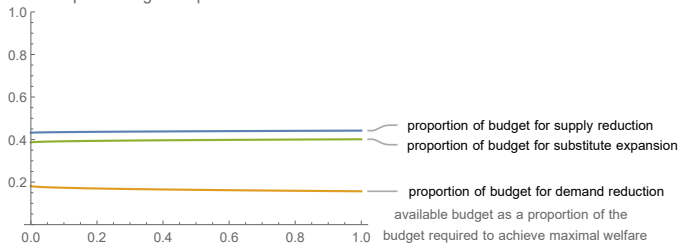
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Constant elasticity specifications

$$\epsilon_x = 1.3, \epsilon_y = 0.85, \epsilon_z = 2.7, \eta = 1.27, \frac{X(0)}{Y(0)} = 0.6, \frac{Z(0)}{Y(0)} = 0.4$$

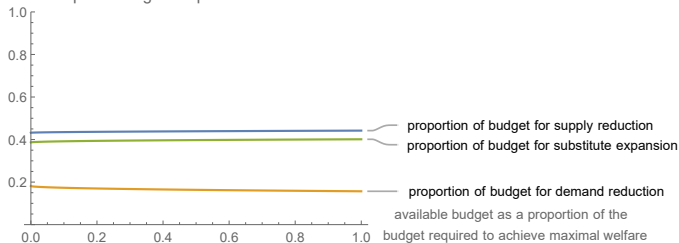
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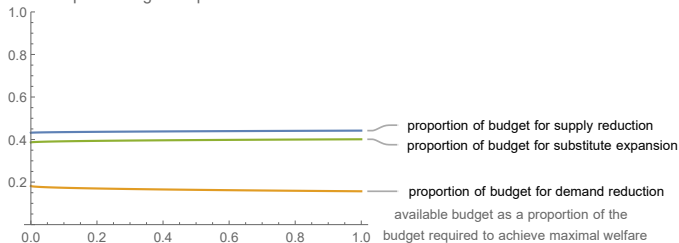
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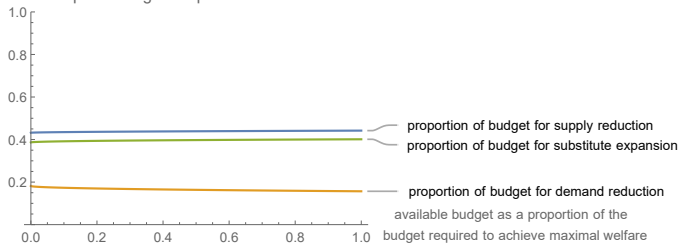
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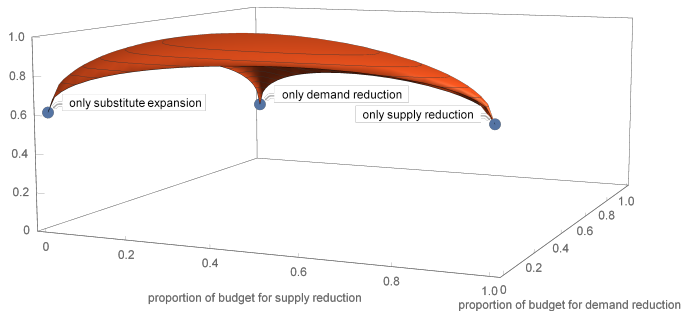
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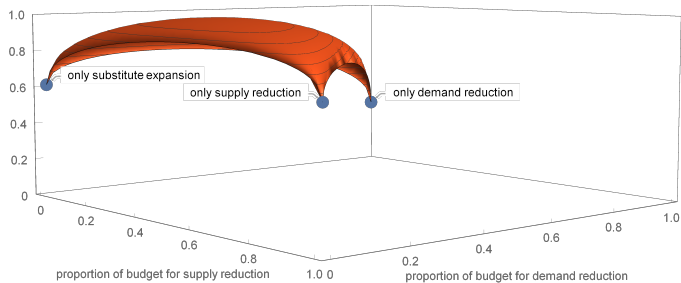
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Global welfare as a function of budget split



Global welfare as a function of budget split



The dynamic model

- y_{it} : country i 's energy use in period t
- z_{it} : country i 's renewable energy production in period t
- x_{it} : country i 's cumulative extraction of coal until the end of period t
- Country i 's utility:
$$U_i = \sum_{t=1}^T \frac{1}{(1+r)^t} (b_{it}(y_{it}) - g_{it}(z_{it}) - (c_{it}(x_{it}) - c_{it}(x_{it-1}))) + f_{it}(x_{it}, y_{it}, z_{it}) + p_t(x_{it} + z_{it} - y_{it}))$$

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$$U_i = \sum_{t=1}^T \frac{1}{(1+r)^t} (b_{it}(y_{it}) - g_{it}(z_{it}) - (c_{it}(x_{it}) - c_{it}(x_{it-1}))) + f_{it}(x_{it}, y_{it}, z_{it}) + p_t(x_{it} + z_{it} - y_{it}))$$

The dynamic model

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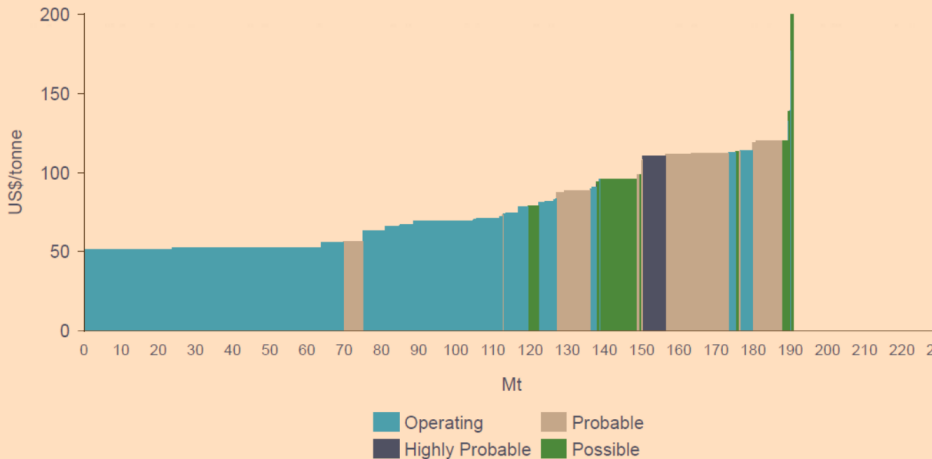
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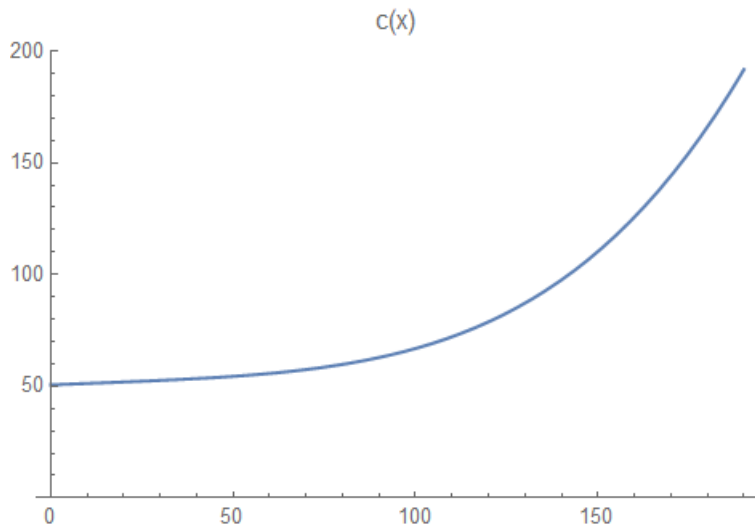
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Empirical estimate of $c(x)$

Base Case Coal Supply Cost Curve – 2035, split by Development Status (2015 prices)



Cubic approximation of the empirical estimate of $c(x)$



The global institution's problem under full commitment and without constraints on borrowing/saving

- Announce path of reward payment schemes $(f_{ixt}(x_{it}), f_{iyt}(y_{it}), f_{izt}(z_{it}))_{i \in I, t \in \{1, \dots, T\}}$ and fully commit to it
- Budget constraint:
$$\sum_{t=1}^T \sum_{i \in I} \frac{1}{(1+r)^t} (f_{ixt}(x_{it}) + f_{iyt}(y_{it}) + f_{izt}(z_{it})) \leq \sum_{t=1}^T \frac{1}{(1+r)^t} F_t$$
- Objective:
$$\sum_{i \in I} U_i - \eta(\sum_{i \in I} x_{iT})$$

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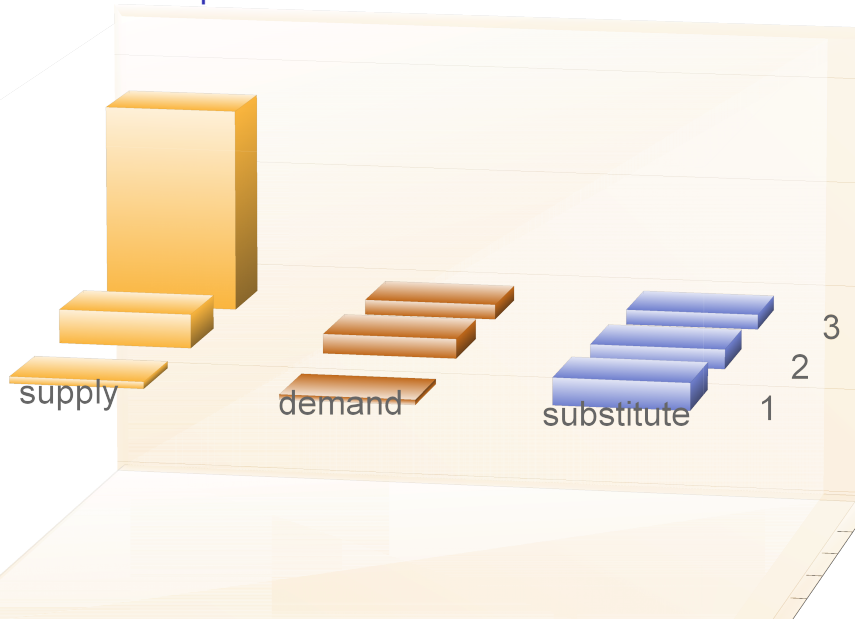
The Dynamic Price Preservation Lemma

Lemma

At the optimal mechanism :

The entire world market price path $(p_t)_{t \in \{1, \dots, T\}}$ is identical to when there is no mechanism.

The global institution's spending path at the optimal mechanism in a 3-period model



The Monotone Mitigation Lemma

Lemma

As the global institution's budget expands:

Coal extraction and consumption decreases at all times and for all countries.

The Monotone Optimal Spending Corollary

Corollary

As the global institution's budget expands:

*The **Global Institution's spending increases** for all approaches and at all periods.*

The Coase-Does-Not-Hold Corollary

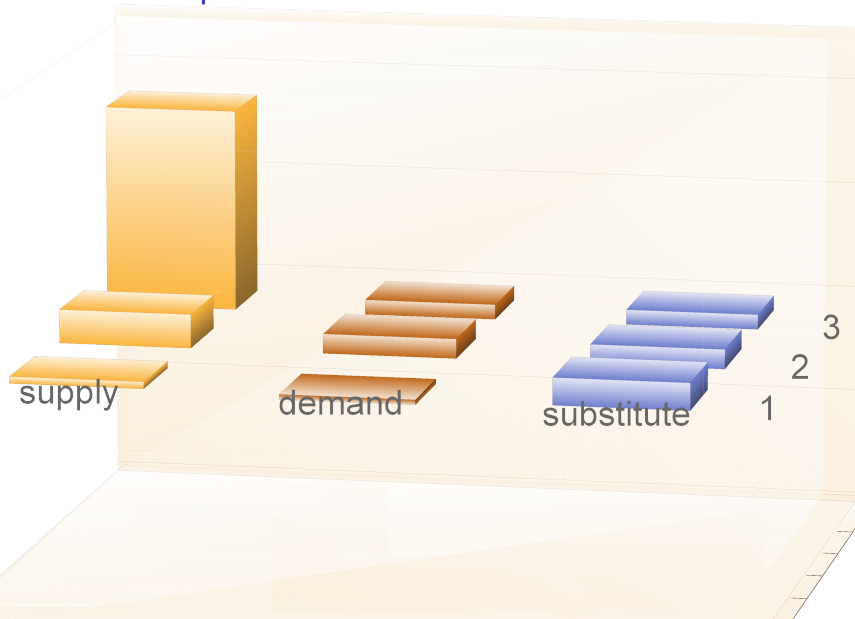
Corollary

Suppose that the global institutions restricts its supply side spending to the last period.

(This would effectively be the case if the global institution were to restrict itself to buying up coal deposits.)

Then the global institution cannot achieve maximal welfare.

The global institution's spending path at the optimal mechanism in a 3-period model



The Time-Inconsistency Corollary

Corollary

Suppose the global institution announces the optimal mechanism under full commitment and all the countries believe it will stick to it. Then the global institution will later have an incentive to renege and spend less on rewarding supply reduction than it announced.

Would it be beneficial to set up global funds reducing fossil fuel supply?

- For oil and gas: net welfare-decreasing
- For coal: net welfare-increasing
 - Could increase emission reduction for a given overall available funding by 25%
- But: Deposit purchase funds alone are insufficient to implement the optimal mechanism on the supply side
- Implementation under informational constraints easier for demand reduction (Edenhofer et al. 2025)
 - ⇒ higher priority to focus on setting up effective global funds for demand reduction

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Appendix slides

A model of the financing game

- player im : A group of coal importers:
 - internalizing s_{im} of global climate change damages
 - making up $2s_{im}$ of coal imports
- player ex : A group of coal exporters:
 - internalizing s_{ex} of global climate change damages
 - making up $2s_{ex}$ of coal exports

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2 architectures

- separated architecture:
 - separate funds for rewarding supply reduction and demand reduction
 - donors can earmark contributions
- unified architecture:
 - unified institution splitting its budget to maximize emissions reductions
 - donors cannot earmark contributions
- **Lemma:** Under both architectures there is a unique Nash Equilibrium.
- **Lemma:** Under the unified architecture, generically, exactly one player contributes.
- **Lemma:** Under the separated architecture both funds receive positive funding.
- **Proposition:** The emissions under the separated architecture are always lower or equal to those under the unified architecture

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Results under linear demand and supply functions

Lemma

Suppose the coal importer and the coal exporter are of equal size: $s_{im} = s_{ex} = s$. Switching from the unified architecture to the separated architecture multiplies the emissions reductions by the following factor:

$$1 + \frac{4 \frac{\alpha}{(e_d + e_s)}}{\eta} \frac{1}{1 - \frac{s}{1-s} \alpha \left(\frac{e_d - e_s}{e_d + e_s} \right)^2}$$

- η = social cost of carbon of coal relative to its price
- α = global coal exports divided by global coal use
- e_d = current price elasticity of demand for coal
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- η = social cost of carbon of coal relative to its price = 1.27
- α = global coal exports divided by global coal use = 0.2
- e_d = current price elasticity of demand for coal = 0.7
- e_s = current price elasticity of supply of coal = 1.3

Results under linear demand and supply functions

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- η = social cost of carbon of **oil** relative to its price = 0.24
- α = global **oil** exports divided by global **oil** use = 0.425
- e_d = current price elasticity of demand for **oil** = 0.5
- e_s = current price elasticity of supply of **oil** = 0.32

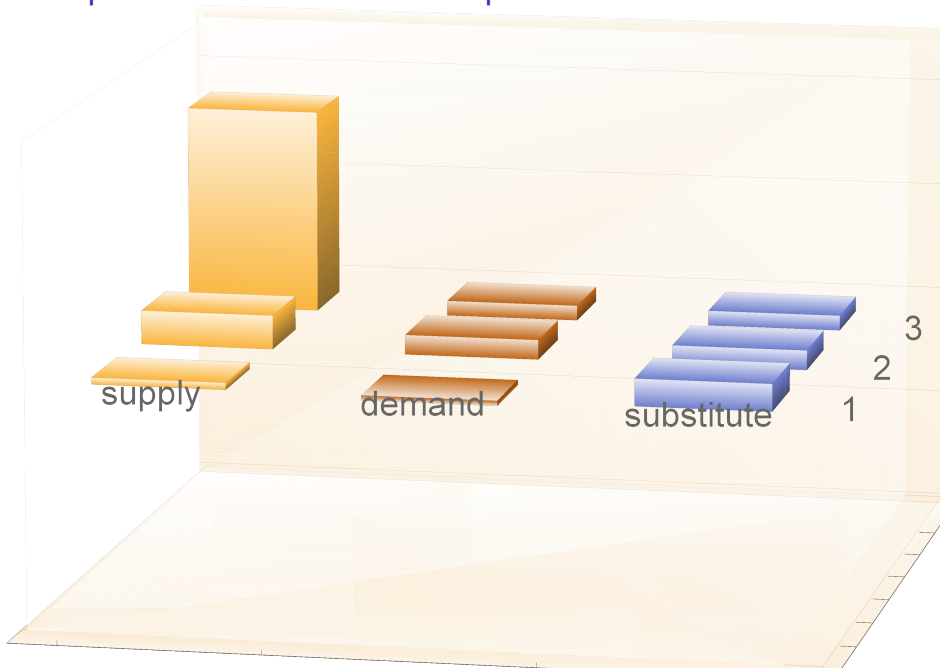
Constrained Efficiency Lemma

Lemma

At the optimal mechanism we have:

The allocation $(x_{it}, y_{it}, z_{it})_{i \in I, t \in \{1, \dots, T\}}$ maximises global welfare amongst all allocations having the same value for climate change damages, $\sum_{i \in I} \eta_i(\sum_{j \in I} x_{j1}, \dots, \sum_{j \in I} x_{jT})$.

The optimal mechanism in a 3 period model



A commitment problem

Corollary

Suppose the global institution announces at time 1 the optimal mechanism assuming it fully commits to it.

Suppose that at time $t > 1$ the global institution announces, to all countries' surprise, a new mechanism that it actually sticks to from then onwards.

Then the new mechanism involves less spending on rewarding supply reduction than the originally announced mechanism.

How severe is the commitment problem?

- From now on suppose that:
 - the global institution cannot commit at all
 - the global institution cannot save or borrow
 - $T=3$
- $x_t(F_1, F_2, F_3) :=$ aggregate cumulative coal extraction in period t
- denote by y_t coal demand in period t

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Additional Funding will always decrease eventual emissions

Proposition

$$\frac{\partial x_3}{\partial F_t} \big|_{(F_1, F_2, F_3)} < 0 \forall (F_1, F_2, F_3) \forall t \in \{1, 2, 3\}$$

The Weak Green Paradox

Proposition

$$\frac{\partial x_1}{\partial F_2} \big|_{(F_1, F_2, F_3)} > 0 \forall (F_1, F_2, F_3)$$

Can climate change damages increase as a result of additional funding for the global institution?

Assumption

$$\eta(x_1, x_2, x_3) = \tilde{\eta}\left(x_1 + \frac{1}{1+r}(x_2 - x_1) + \frac{1}{(1+r)^2}(x_3 - x_2)\right)$$

- By preceding propositions, increasing F_2 will:
 - increase x_1
 - decrease x_2 and x_3

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Necessary and Sufficient Conditions for the Strong Green Paradox to not arise for Small Budgets

Proposition

Denote $e_{xt} := \frac{dx_t^*}{dp_t} \frac{p_t}{x_t^*}$, $e_{yt} := -\frac{dy_t^*}{dp_t} \frac{p_t}{y_t^*}$ and

$$a := e_{y1} \frac{1}{\left(\frac{x_3 - x_2}{x_1}\right) \left(1 + \frac{e_{y1}}{e_{x1}}\right) \left(1 + \frac{e_{y2}}{e_{x2}} \left(\frac{x_2 - x_1}{x_2}\right)\right) \frac{p_1(1+r)}{p_3} + \frac{e_{y1}}{e_{x2}} \frac{x_1}{x_2} \frac{p_2}{p_3}}$$

Then the following condition is sufficient for the Strong Green Paradox to not occur for small budgets:

$$e_{y3} \geq a$$

Moreover, if $e_{y3} < a$ then the following condition is necessary and sufficient for the Strong Green Paradox to not occur for small budgets:

$$e_{x3} \geq e_{y1} \frac{\frac{a - e_{y3}}{a} r}{\frac{x_3}{x_2} \left(1 + \frac{e_{y1}}{e_{x1}}\right) \left(\frac{x_2 - x_1}{x_1} + \frac{x_2}{x_1} \frac{e_{y2}}{e_{x2}}\right) \frac{p_1(1+r)^2}{p_3} + \frac{e_{y1}}{e_{x2}} \frac{p_2(1+r)}{p_3}}$$

Necessary and Sufficient Conditions for the Strong Green Paradox to not arise for Small Budgets

Proposition

Suppose that extraction costs do not change over time, so that we can denote them simply by $c(x)$.

Then the Strong Green Paradox occurs for small budgets in the three period model iff $h := g_1 g_2 + g_3 + g_4 (g_5 + g_6) < 0$

with the following definitions:

$$e_{yt} := - \frac{dy_t^*}{dp_t} \frac{p_t}{y_t^*}$$

$$g_1 := rc''(x_2) (e_{y3} r(x_3 - x_2) c''(x_3) + (1+r) c'(x_3))$$

$$g_2 := (e_{y1} x_1 + e_{y2} (x_2 - x_1)) (rc'(x_2) + c'(x_3)) - e_{y2} r(1+r) (x_1 - x_2) c'(x_1)$$

$$g_3 := e_{y1} r(1 +$$

$$r) x_1 c''(x_1) (e_{y3} r(x_3 - x_2) c''(x_3) + (r+1) c'(x_3)) (e_{y2} r(x_2 - x_1) c''(x_2) + rc'(x_2) + c'(x_3))$$

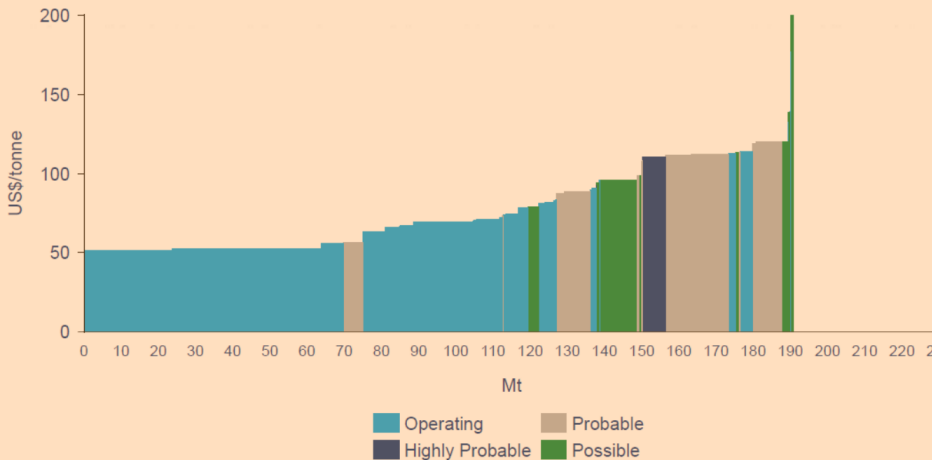
$$g_4 := rc'(x_2) + c'(x_3)$$

$$g_5 := (1+r) c'(x_3) (r(1+r) c'(x_1) + rc'(x_2) + c'(x_3))$$

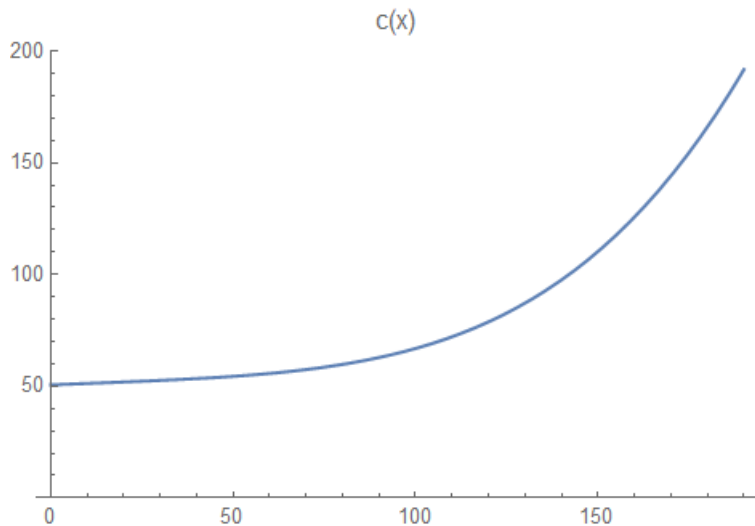
$$g_6 := rc''(x_3) (e_{y3} (x_3 - x_2) (r(1+r) c'(x_1) + rc'(x_2) + c'(x_3)) - e_{y1} (1+r) x_1 c'(x_3))$$

Empirical estimate of $c(x)$

Base Case Coal Supply Cost Curve – 2035, split by Development Status (2015 prices)



Cubic approximation of the empirical estimate of $c(x)$



Corollary

Suppose that the third period extraction is not more than the second period extraction. Then the Strong Green Paradox can only occur for small budgets if $\frac{e_{y1}}{e_{y2}} > 15.55$ or $\frac{e_{y1}}{e_{y3}} > 15.55$.

good with global externalities	substitute	international institutions focused on supply	international institutions focused on demand	international institutions focused on substitute
fossil fuels	renewables, nuclear		CDM, GCF	CDM, IAEA's nuclear
goods produced on previously forested land (soy, palm oil...)	same goods produced on non-forested land	REDD+		
over-fished fish	bivalve aquaculture	WTO negotiations on reducing fishery subsidies		
drugs and vaccines for infectious disease control	single product treatments increasing risk of drug resistance		Global Alliance for Vaccines and Immunization Global Fund for AIDS, TB and Malaria	
goods for pandemic surveillance			International Health Regulations	
PPE for pandemic preparedness				